Time-Series Analysis and Forecasting of Monthly U.S. Retail Sales

Introduction

Why Retail Sales Forecasting Matters?

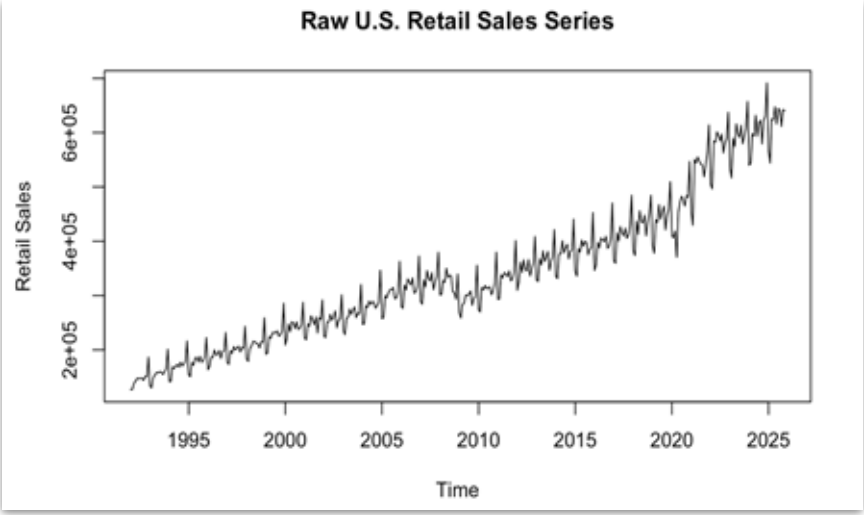
- Retail sales reflect consumer spending and economic activity
- Data exhibit trend and strong seasonal patterns
- Forecasting helps support business planning and policy decisions

Project Objective

- Analyze U.S. monthly retail sales time series
- Compare forecasting models from different categories
- Identify the model with best predictive performance

Dataset Description

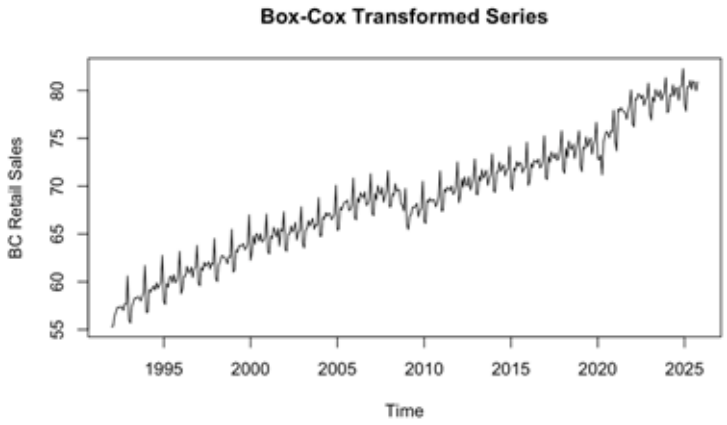
- **Data Source:** Federal Reserve Economic Data (FRED)
- **Series:** Monthly U.S. Retail Sales
- **Series ID:** MRTSSM44000USN
- **Time Coverage:** January 1992 – October 2025
- **Frequency:** Monthly observations
- **Sample Size:** 406 observations
- **Measurement:** Retail sales reported in millions of U.S. dollars

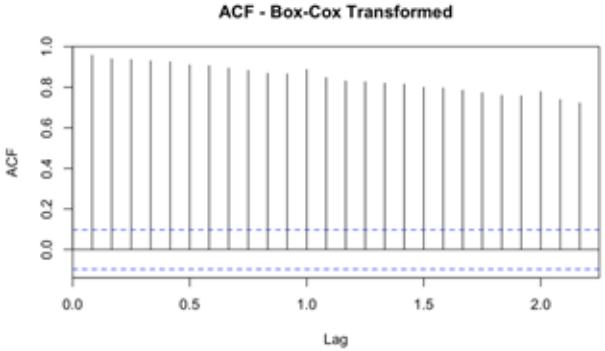


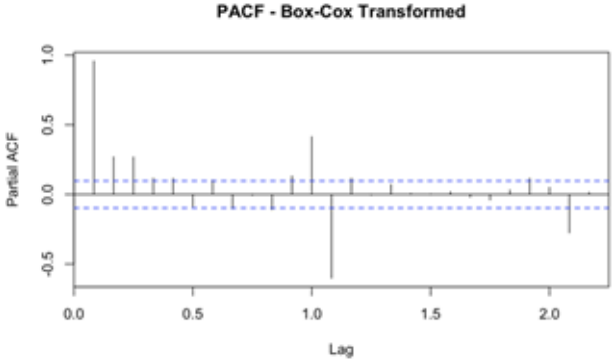
Exploratory Data Analysis

Step 1: Box-Cox transformation

- Variance becomes constant across time
- ACF shows extremely slow decay
- PACF shows significant spikes at seasonal lags
- So differencing is needed



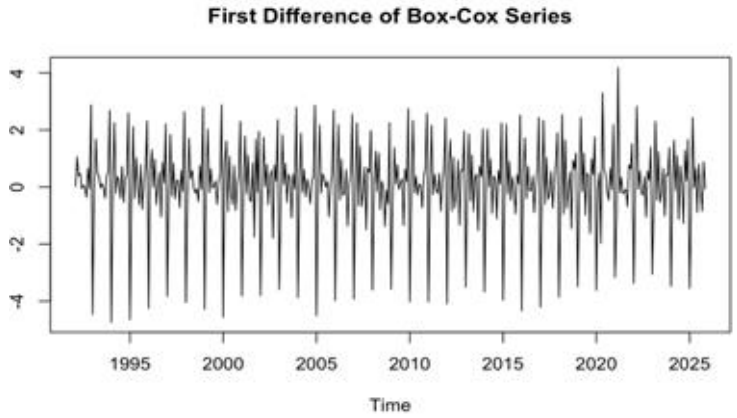


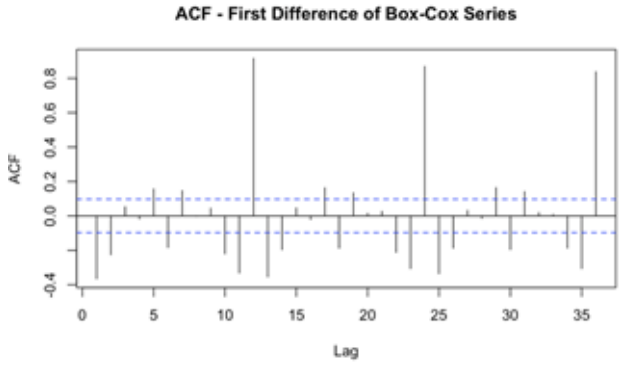


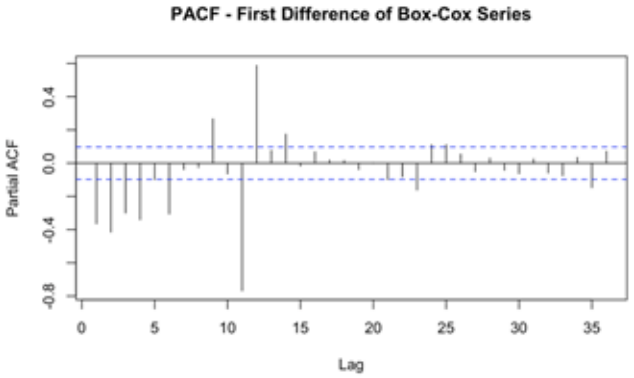
Exploratory Data Analysis

Step 2: First-Order Differencing

- Trend is removed
- Seasonality still exist
- ACF and PACF both have very high spikes (around 0.8) at seasonal lags
- So seasonal differencing is needed



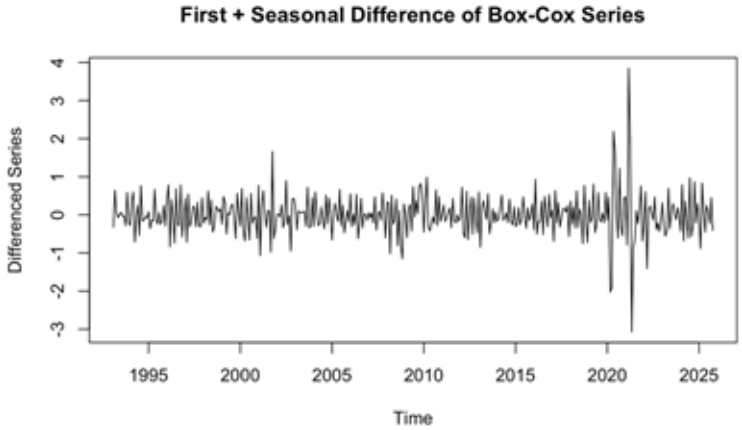


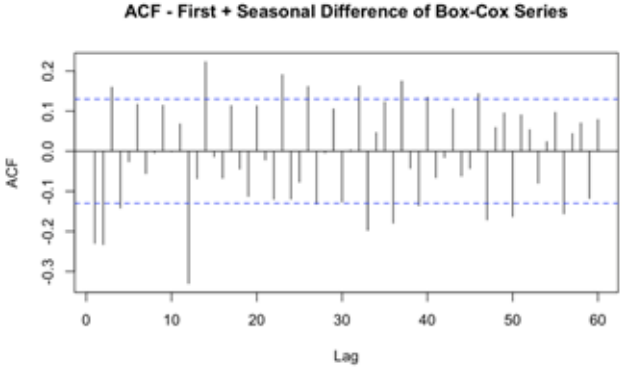


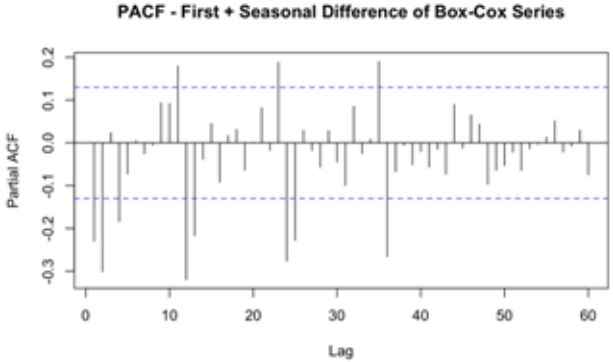
Exploratory Data Analysis

Step 3: Seasonal Differencing

- Time plot resembles white noise (except one structural break)
- Magnitude of spikes in ACF and PACF largely drops (from 0.8 to 0.3) and disappear after 3 to 4 seasons
- Most importantly, it now passes both ADF and KPSS stationarity tests







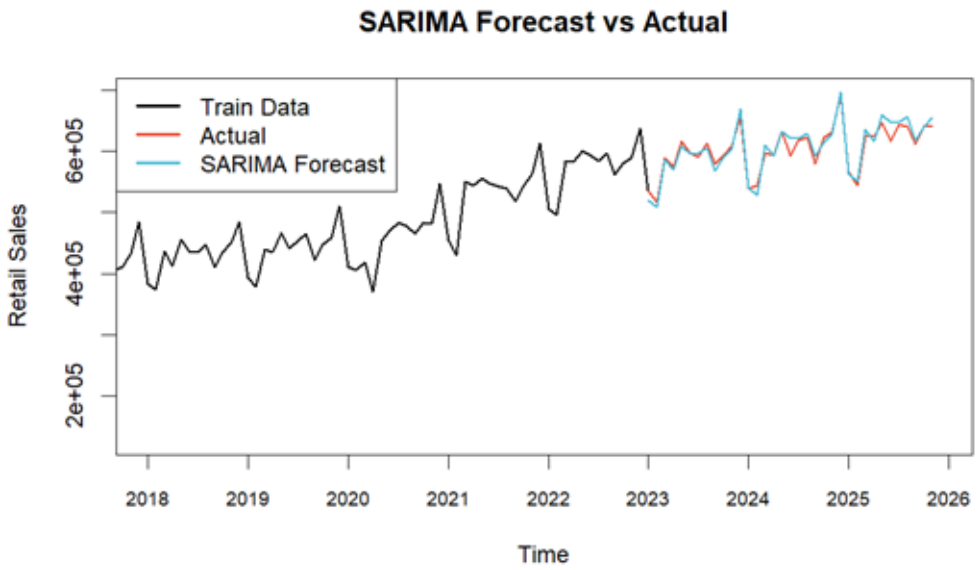
SARIMA

Model Idea

- Captures trend, autocorrelation, and seasonality
- Uses monthly retail sales structure for forecasting
- Advanced time series model compared with baseline

Model Specification

- Fitted on log-transformed training series
- Selected model: ARIMA(3,0,1)(0,1,1)[12] with drift



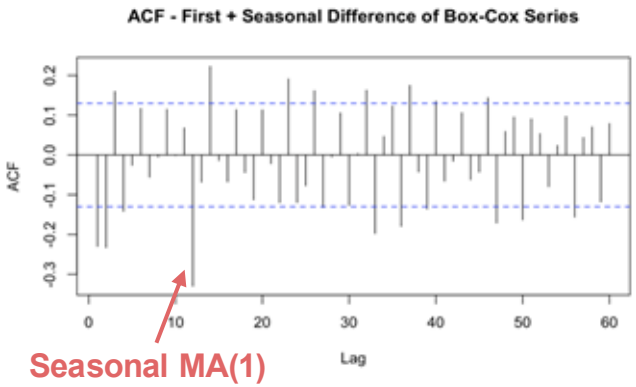
Performance	
RMSE	11024.31
MAE	8603.78
MAPE	1.43%

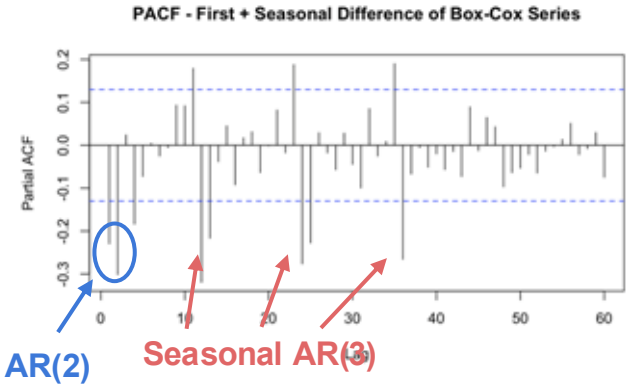
SARIMA

However, **autocorrelation** remains in the residuals:

```
        Ljung-Box test

data:  Residuals from ARIMA(3,0,1)(0,1,1)[12] with drift
Q* = 60.434, df = 19, p-value = 3.304e-06
```

To address this, we moved to models with higher orders to capture remaining autocorrelation.





Based on the patterns of ACF and PACF plots, we started with ARIMA(2,1,0)(3,1,1)_12. However, its Ljung-Box p-value is 0.001, still smaller than the significance level of 0.01.

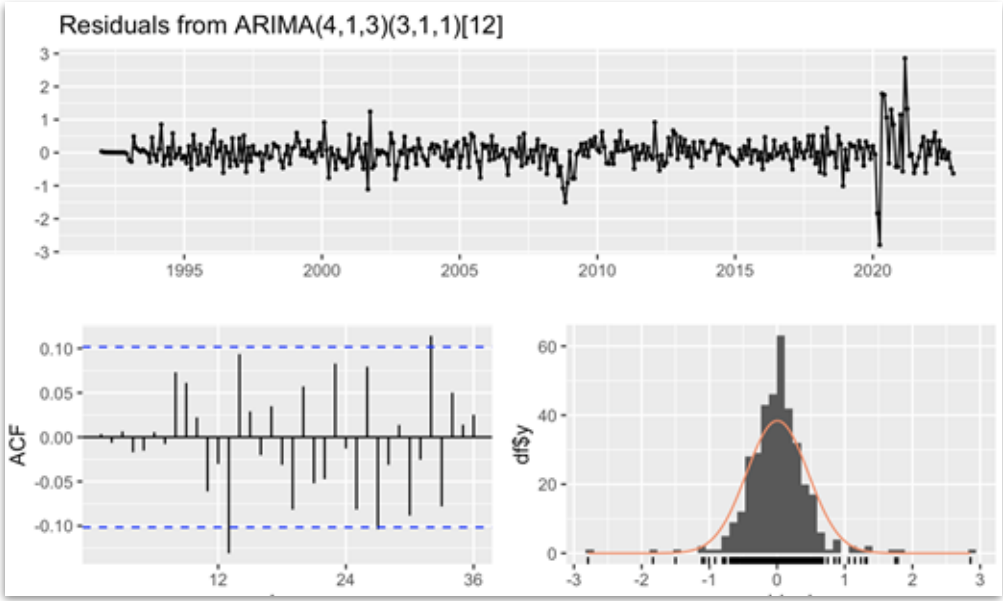
SARIMA

We then experimented with models with other combinations of orders and compared them:

Model <chr>	RMSE <dbl>	MAPE <dbl>	AICc <dbl>	BIC <dbl>	LjungBox_pvalue <dbl>
ARIMA(4,1,3)(3,1,1)	11542.15	1.477193	508.6460	554.3441	0.018
ARIMA(1,1,1)(3,1,1)	11862.33	1.486232	507.1143	533.9784	0.003
ARIMA(2,1,0)(3,1,1)	11983.43	1.525518	507.6105	534.4747	0.001
ARIMA(2,1,0)(3,1,1)	11983.43	1.525518	507.6105	534.4747	0.001
auto_sarima	12040.03	1.707865	525.1337	552.0183	0.000
auto_sarima_fixed_d	14947.01	2.027907	513.7834	536.8447	0.000
ARIMA(2,1,2)(3,1,2)	15264.37	1.876989	499.2891	537.4901	0.085
ARIMA(4,1,3)(3,1,2)	15444.51	1.903537	501.6046	551.0328	0.128
ARIMA(4,1,2)(3,1,2)	15946.50	1.955749	499.1543	544.8524	0.161
ARIMA(3,1,3)(3,1,1)	18565.26	2.381987	482.5260	524.4818	0.074

ARIMA(4,1,3)(3,1,1) wins, with the lowest error, low AICc, and Ljung-Box p-value > 0.01 significance level.

SARIMA - Residual Diagnostics



- Time Plot: Residuals resembles white noise (except the 2020 structural break).
- ACF Plot: Most bars fall within the 95% confidence interval, with minor significant spikes (≈ 0.1).
- Histogram: Distribution is close to normal, though with heavier tails (likely driven by the 2020 structural break).
- Ljung-Box p-value: 0.018 > 0.01.

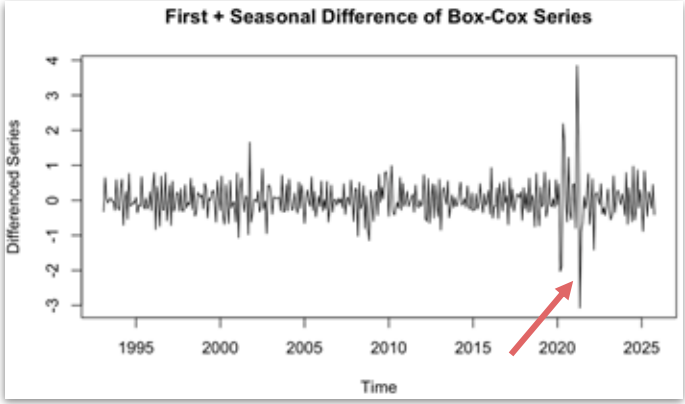
Regression with ARMA Errors

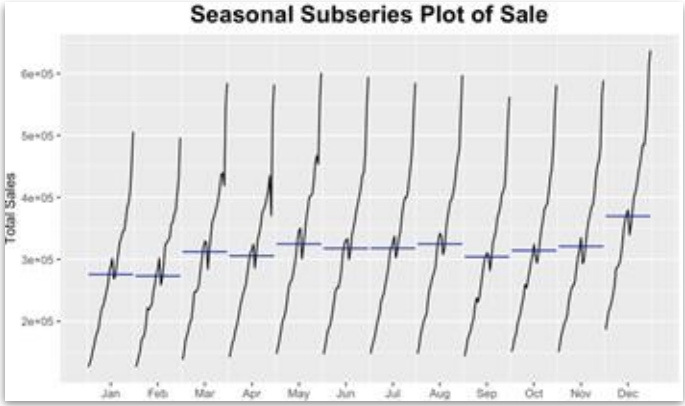
Volatility increases significantly during the COVID period.

=> Include a COVID-period indicator variable.

Retail sales spike every December and drop sharply every January and February.

=> Include month as a dummy variable.





Regression with ARMA Errors (Auto)

With covid indicator variable:

```
Regression with ARIMA(3,0,1)(0,1,1)[12] errors
Box Cox transformation: lambda= 0.2339066

Coefficients:
         ar1     ar2     ar3     ma1     sma1   drift     xreg
      0.2370  0.3120  0.3883  0.3604  -0.8268  0.0682  -0.5105
s.e.  0.1304  0.1035  0.0556  0.1407   0.0358  0.0088   0.3281

sigma^2 = 0.2357:  log likelihood = -254.23
AIC=524.46   AICc=524.87   BIC=555.55
```

The covid term is not significant (|t| < 1.96).

Regression with ARMA Errors (Auto)

With month dummy variable:

```
Regression with ARIMA(0,1,2)(2,0,0)[12] errors
Box Cox transformation: lambda= 0.2339066

Coefficients:
         ma1      ma2    sar1     sar2   drift      Jan      Feb      Mar      Apr
      -0.410  -0.0912  0.1479  -0.1204  0.0703  -5.0228  -5.2242  -2.8397  -3.2762
s.e.   0.056   0.0594  0.0567   0.0564  0.0126   0.0943   0.1079   0.1127   0.1162
          May      Jun      Jul      Aug      Sep      Oct      Nov
      -2.1945  -2.6736  -2.7475  -2.3774  -3.7077  -3.1488  -2.8351
s.e.   0.1179   0.1185   0.1178   0.1157   0.1123   0.1073   0.0934

sigma^2 = 0.2296:  log likelihood = -245.72
AIC=525.44   AICc=527.17   BIC=592.01
```

The month terms are all significant (t > 1.96).

Regression with ARMA Errors (Manual)

However, autocorrelation remains in the residuals:

```
        Ljung-Box test

data:  Residuals from Regression with ARIMA(0,1,2)(2,0,0)
[12] errors
Q* = 70.742, df = 20, p-value = 1.377e-07
```

To address this, we moved to models with higher orders to capture remaining autocorrelation.

Model	RMSE	MAPE	AICc	BIC	LjungBox_pvalue
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
reg_arma(2,1,4)(3,0,1)	13543.92	1.972061	492.9992	576.2476	0.011
reg_arma(3,1,3)(2,0,3)	15135.42	2.231111	491.7350	578.6261	0.022
reg_arma(2,1,3)(3,0,1)	15875.54	2.366534	492.4580	572.0507	0.013
reg_arma(2,1,3)(3,0,2)	17405.12	2.638909	500.0747	583.3231	0.047
reg_arma(3,1,2)(2,1,3)	19081.40	2.530506	506.9828	589.4040	0.085
reg_arma(2,1,3)(2,1,1)	19616.32	2.670867	499.1728	574.3541	0.028
reg_arma(2,1,1)(2,0,3)	41251.34	5.907513	544.8386	620.7627	0.000

reg_arma(2,1,4)(3,0,1) wins, with lower errors and AICc/BIC and p-value > 0.01.

ETS(M, A, A) Model — Holt-Winters

Multiplicative Error · Additive Trend · Additive Seasonality

Smoothing Parameters

α (alpha)	0.5346
β (beta)	1e-04
γ (gamma)	0.0765

R Output

```
ETS(M,A,A)

Call:
ets(y = train_boxcox)

  Smoothing parameters:
    alpha = 0.5346
    beta  = 1e-04
    gamma = 0.0765

  Initial states:
    l = 64.3671
    b = 0.0716
    s = 3.2769 0.1386 -0.1054 -0.661 0.4815 0.2
           0.287 0.7168 -0.1376 0.0157 -2.1124 -2.1002

  sigma:  0.0061

     AIC     AICc      BIC
1662.671 1664.399 1729.292

Training set error measures:
                        ME     RMSE     MAE          MPE      MAPE     MASE       ACF1
Training set -0.0009641154 0.481312 0.33945 -0.003491848 0.4356808 0.320203 0.03497747
```

Model Evaluation — Test Set (2023–2025)

RMSE

10,019.5

Root Mean Squared Error
(on original scale)

MAPE

1.34%

Mean Absolute Percentage Error
(excellent: < 5% is strong)

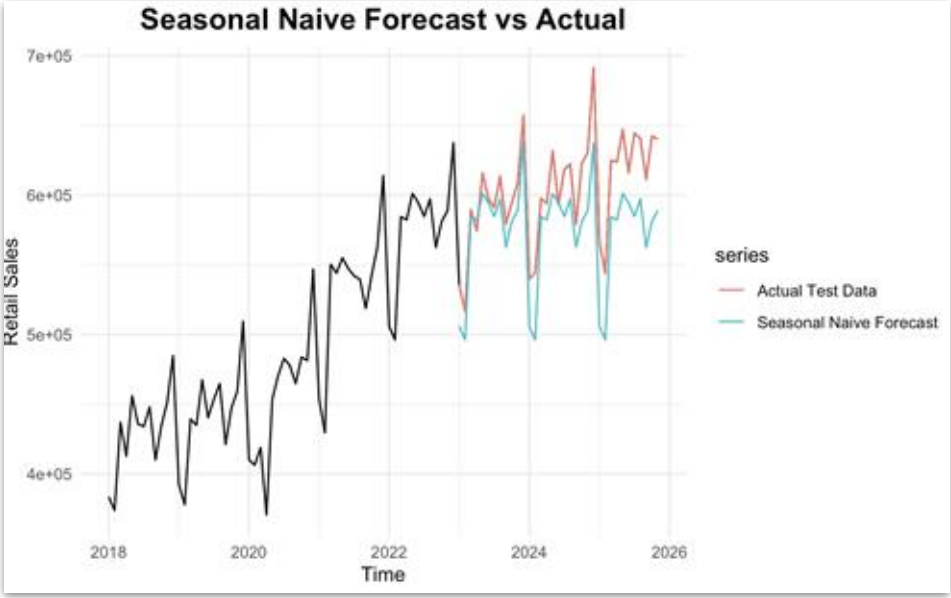
✓ ETS(M,A,A) forecast was generated on Box-Cox transformed training data, then back-transformed for evaluation.
MAPE of 1.34% demonstrates strong predictive accuracy relative to U.S. retail sales scale.

Baseline Model for Comparison: Seasonal Naive

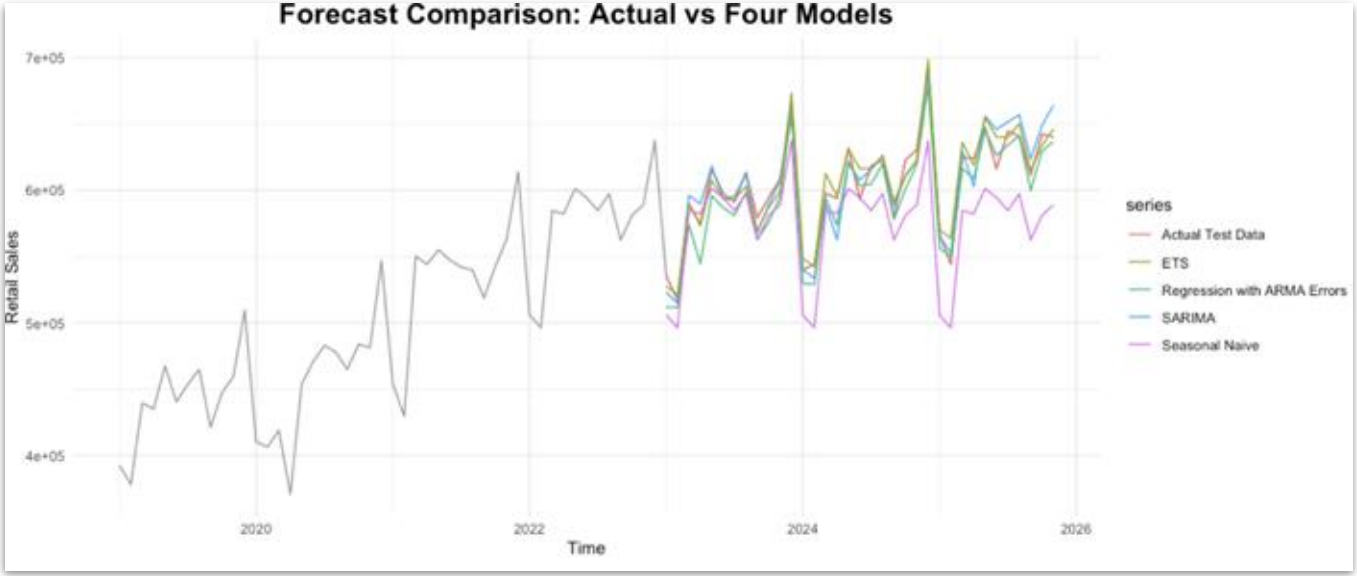
Model Idea

- Because our data show strong, consistent yearly seasonal pattern
- Forecast equals value from same month last year
- Simple benchmark model

Performance	
RMSE	34825.01
MAE	29926.57
MAPE	4.93%

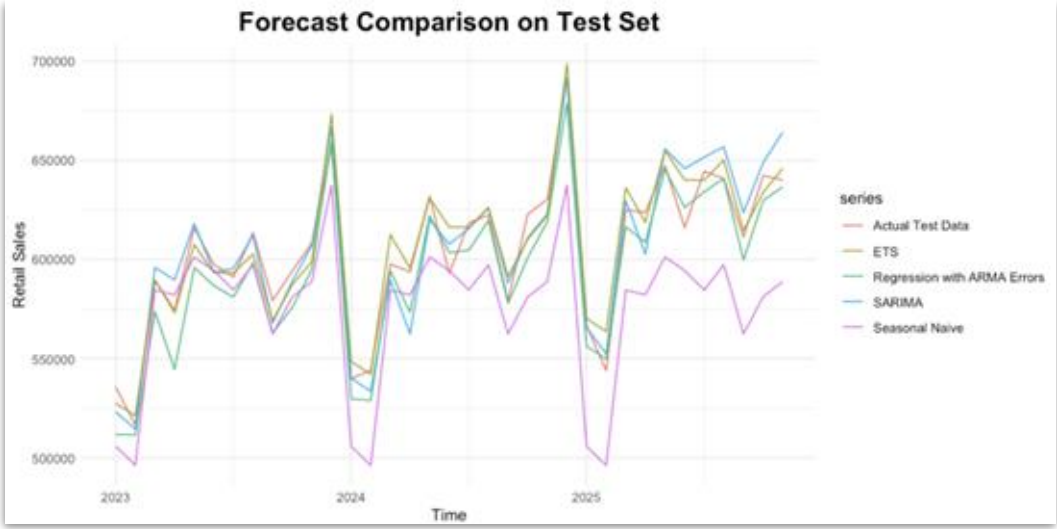


Forecast Performance Comparison across Models



- Seasonal naive underestimates the series because it cannot capture the upward trend.
- ETS, SARIMA, and Regression-ARMA track the actual data more closely.

Forecast Performance Comparison across Models

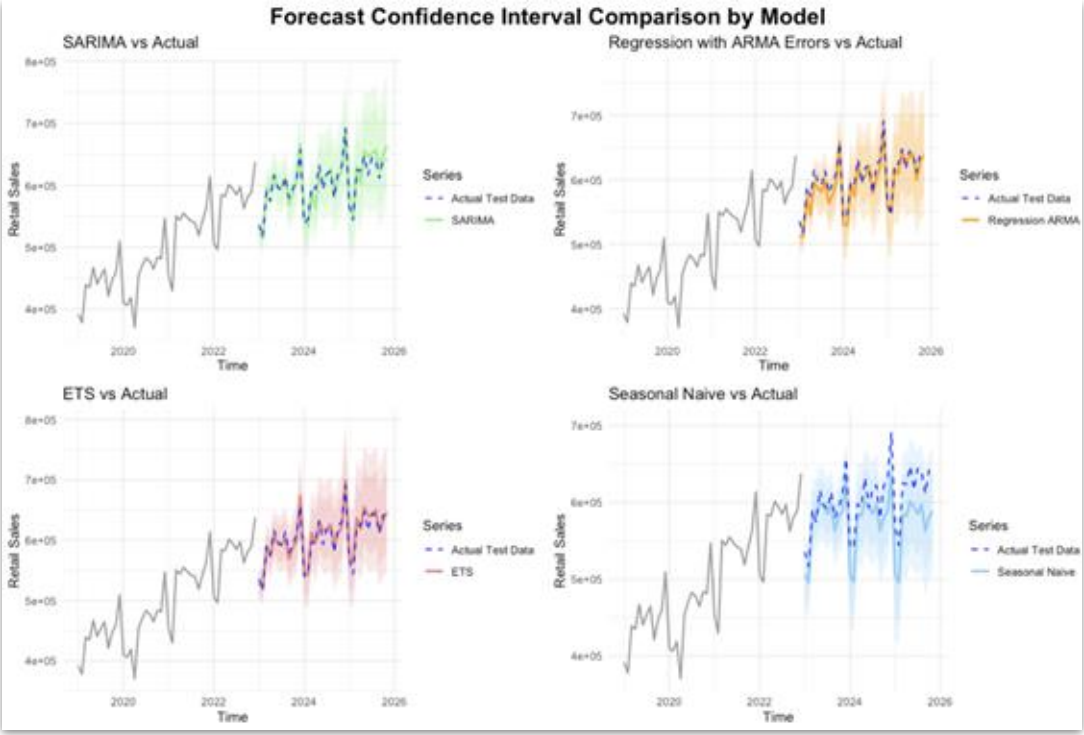


- Zoomed view of the test period to highlight differences between models.
- ETS shows the best alignment with the observed data.

Model <chr>	RMSE <dbl>	MAPE <dbl>	LjungBox_pvalue <dbl>
ETS	10019.50	1.341733	0.000
SARIMA	12082.35	1.543389	0.018
Regression with ARMA Errors	13375.12	1.933383	0.011
Seasonal Naive	34825.01	4.926572	0.000

Forecast Performance Comparison across Models

- Shaded areas represent forecast confidence intervals.
- Advanced models provide reasonable uncertainty estimates.

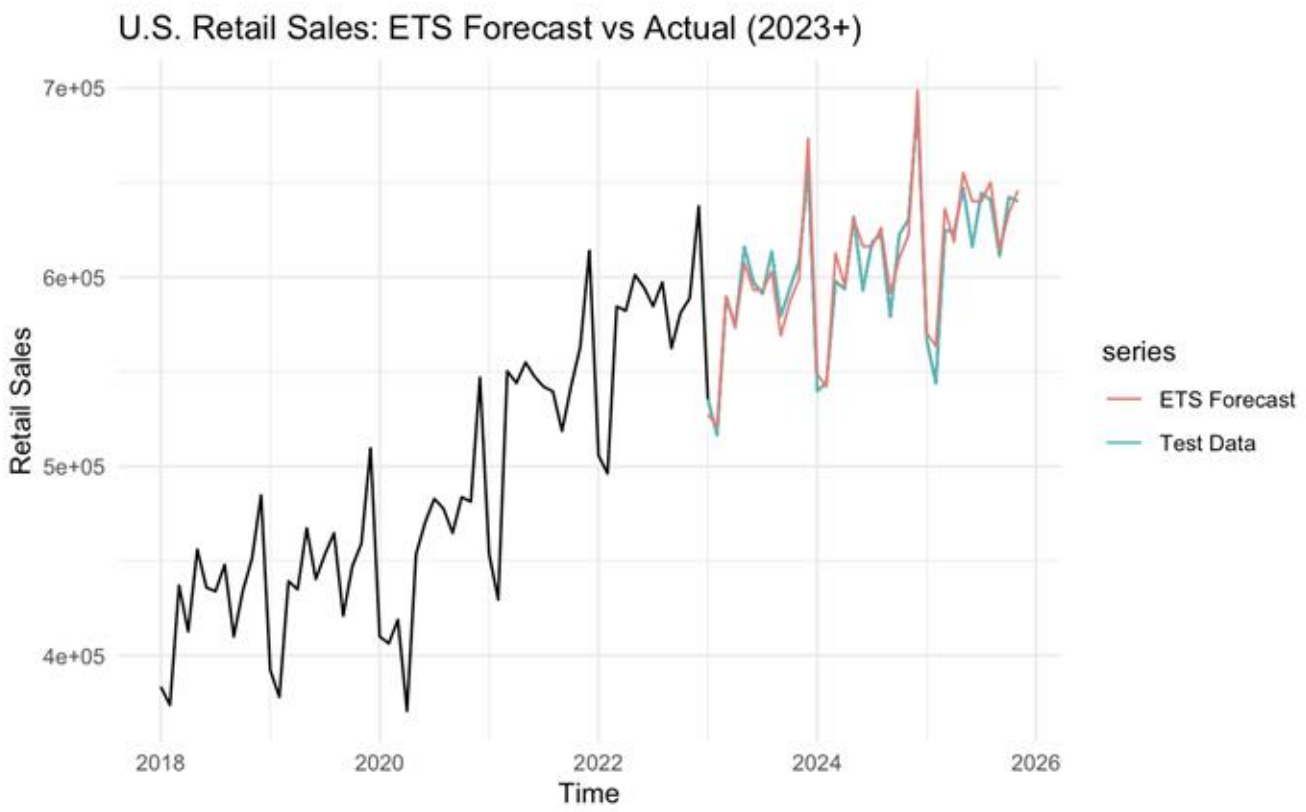


Future Work

Structural Break from COVID-19

- The COVID-19 outbreak caused a large increase in variance, indicating a structural break in volatility rather than in the mean.
- This suggests the time-series properties changed during the pandemic period.
- One possible approach is to train the model using pre-COVID data and test it on post-COVID data to evaluate generalization.
- We can then compare its performance with a model trained only on post-COVID data.
- This comparison helps determine whether models fitted after the structural break provide better forecasts.

Appendix

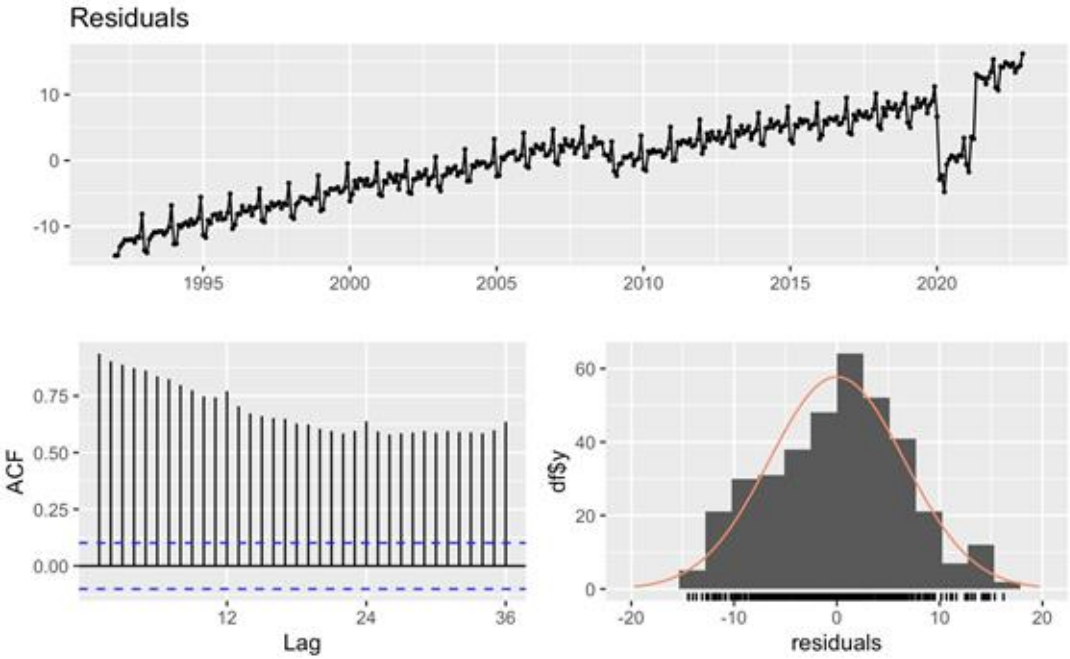

U.S. Retail Sales: ETS Forecast vs Actual (2023+)
7e+05
6e+05
5e+05
4e+05
Retail Sales
2018
2020
2022
2024
2026
Time
series
ETS Forecast
Test Data

```
        Ljung-Box test

data:  Residuals from ARIMA(4,1,3)(3,1,1)[12]
Q* = 25.826, df = 13, p-value = 0.01794

Model df: 11.   Total lags used: 24
```

```
        Ljung-Box test

data:  Residuals from Regression with ARIMA(2,1,4)(3,0,1)
[12] errors
Q* = 28.869, df = 14, p-value = 0.01088

Model df: 10.   Total lags used: 24
```

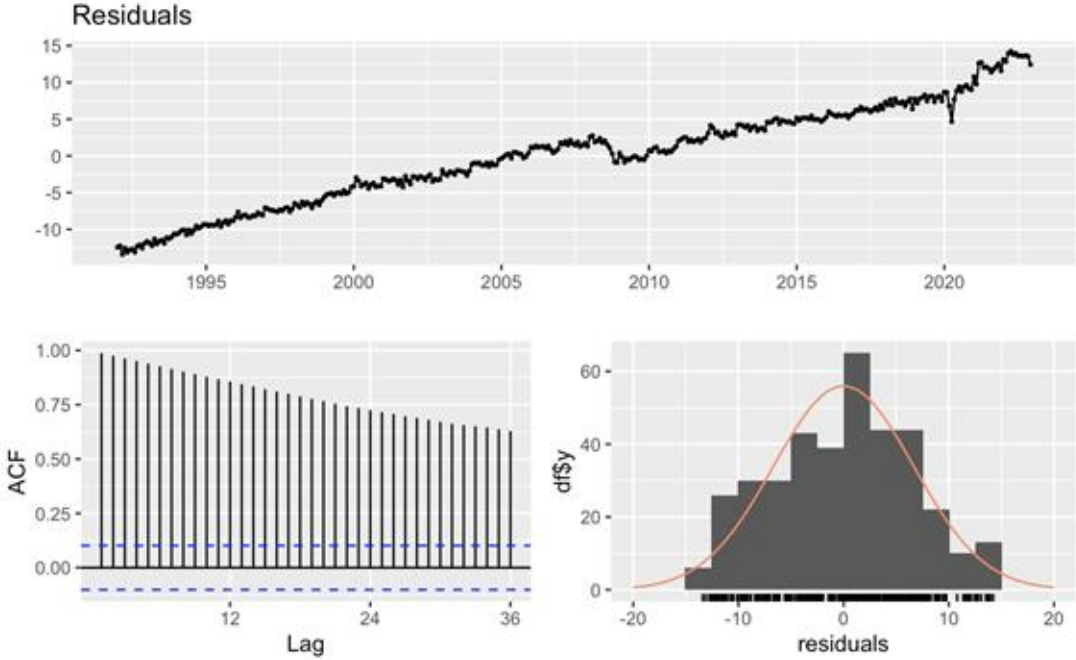
Residual Diagnostics for Ordinary Regression with Covid:

```
        Ljung-Box test

data:  Residuals
Q* = 5062.6, df = 24, p-value < 2.2e-16

Model df: 0.   Total lags used: 24
```



Residual Diagnostics for Ordinary Regression with Month:

```
        Ljung-Box test

data:  Residuals
Q* = 6785.4, df = 24, p-value < 2.2e-16

Model df: 0.   Total lags used: 24
```



Thanks!